

Aditya Kulkarni

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PROFESSIONAL SUMMARY

MS CS candidate (NC State, May 2026) with full-stack delivery across **TypeScript, Node.js, React, PostgreSQL, and REST APIs**. Shipped 10+ production backend features at Isomer AI (394 commits, 7 deployments) and built finance-domain applications including a GenAI finance app (PerCent, Top 10 of 100+ teams) and a compliance-ready fintech RAG system (Glance AI). Java in toolset; learning Angular -- strong React/TypeScript foundation transfers directly. Comfortable owning features end-to-end, from database schema through REST API to UI. H-1B visa sponsorship required.

CORE COMPETENCIES

Full-Stack Development

REST APIs

TypeScript & React

Financial Domain

PostgreSQL & Databases

Cloud (AWS / GCP)

Java & Node.js

Production Software Delivery

WORK EXPERIENCE

Isomer AI May 2025 - Present

Product Engineer Intern

- Built 10+ full-stack features for a production claims automation platform: **REST APIs**, role-based access control, real-time event pipelines, and document processing pipelines (PDF, email, S3) using **TypeScript, Node.js, Express, PostgreSQL, Prisma, and Redis -- 394 commits, 7 production deployments**
- Integrated Temporal.io workflow orchestration, multi-model AI agent tooling across 3 providers, Auth0, Docker, and Sentry observability; owned feature delivery end-to-end from schema design through deployment

Mphasis (HP Company) Jun 2024 - Aug 2024

Software Developer Intern

- Built 5+ AI-driven workflows including a health claim automation tool using OCR, NLP, and ML -- improved claim accuracy by **25%**; automated deployments on **AWS and GCP** with Grafana monitoring dashboards

NC State University Apr 2025 - Present

Graduate Research Assistant

- Designed and deployed a 500GB cloud data platform on **Google Cloud Run** (100+ researchers worldwide); applied time-series analysis and hyperparameter tuning to improve model accuracy 12%

Capgemini Jun 2023 - Aug 2023

AI/ML Intern

- Led end-to-end SDLC for an automated support ticket analysis system; implemented distributed pipelines with **Kafka + Spark** -- reduced latency 20% and processing time 15%

PROJECTS

PerCent -- GenAI Finance App HackNC24 Finalist · Top 10 of 100+ teams, Nov 2024

Built a full-stack GenAI finance application with goal tracking, expense logging, and an AI financial advisor chatbot. Designed a RAG-based chatbot using Hugging Face and Qwen-14B-Chat, reducing manual financial research by **30%**. Delivered a complete working product within a 48-hour hackathon.

Python · React · RAG · Hugging Face · Selenium · REST APIs

Glance AI -- Fintech Compliance RAG System Independent, 2025

Built a compliance-ready RAG system for fintech/healthcare environments with **per-tenant data isolation** at the vector retrieval layer, preventing cross-tenant context leakage. Designed deterministic output architecture ensuring factual correctness for regulated financial data environments.

Python · Vector Databases · RAG · LangChain · LLMs

AlgoPulse -- Full-Stack Event-Driven Platform Independent, 2025

Architected a full-stack platform in **TypeScript + React + Node.js**; identified API-polling bottleneck under load and re-architected to event-driven streaming with Temporal.io workflow state machines and Kafka message brokering -- sustained **2,000 async requests/second** with zero message loss.

TypeScript · React · Node.js · Temporal.io · Kafka · Supabase Edge Functions

Sanjeevani -- Full-Stack Health App Top 1% of 1,100+ submissions, 2023

Developed a full-stack ML-powered health application achieving 87% prediction accuracy using real-time sensor data and classification models. Shortlisted Top 1% (12 of 1,100+) for UI innovation and predictive insights; built with **React** and a REST backend.

React · Python · ML · REST APIs · Full-Stack

EDUCATION

MS Computer Science NC State University

Aug 2024 - May 2026

Operating Systems, Algorithms, Software Engineering, Data Science, Distributed Systems, Object-Oriented Development

B.Tech Computer Science Mumbai University

Aug 2020 - May 2024

CGPA 9.32/10 · Big Data Infrastructure, DS & Algorithms, Machine Learning, Software Engineering, Blockchain

SKILLS

Languages: TypeScript, JavaScript, Python, Java, SQL, C, R · Angular (learning)

Frontend & Backend: React, Node.js, Express, REST APIs, PostgreSQL, Prisma, Redis, Docker, CI/CD

Cloud & DevOps: AWS, GCP, Kubernetes, Grafana, Sentry, Cloud Run, Terraform, GitHub Actions

Finance Domain: PerCent (GenAI finance app), Glance AI (fintech compliance), Isomer AI (insurance claims automation)

Awards: Top 10 in 6+ national hackathons · Google Developer Student Club Lead (500+ students) · IEEE Published